Table 6. AES features

Modes or features[1]	AES
ECB, CBC chaining	X
CTR, CCM, GCM chaining	X
AES 128-bit ECB encryption in cycles	51
DHUK and BHK key selection	-
Resistant to side-channel attacks	-
Shared key between SAES and AES	X
Key sizes in bits	128/256

1. *X = supported.*

3.20 HASH processor (HASH)

The hash processor is a fully compliant implementation of the secure hash algorithm (SHA-1, SHA-2 family) and the HMAC (keyed-hash message authentication code) algorithm. HMAC is suitable for applications requiring message authentication.

The hash processor computes FIPS (Federal Information Processing Standards) approved digests of lengths of 160, 224, and 256 bits for messages of any length less than 2 × 64 bits (for SHA-1, SHA-224, and SHA-256).

The HASH main features are:

- Suitable for data authentication applications, compliant with:
  - Federal Information Processing Standards Publication FIPS PUB 180-4, Secure Hash Standard (SHA-1 and SHA-2 family)
  - Federal Information Processing Standards Publication FIPS PUB 186-4, Digital Signature Standard (DSS)
  - Internet Engineering Task Force (IETF) Request For Comments RFC 2104, HMAC: Keyed-Hashing for Message Authentication and Federal Information Processing Standards Publication FIPS PUB 198-1, The Keyed-Hash Message Authentication Code (HMAC)
- Fast computation of SHA-1, SHA2-224, SHA2-256:
  - 82 (respectively 66) clock cycles for processing one 512-bit block of data using SHA-1 (respectively SHA-256) algorithm
- Support for HMAC mode with all supported algorithms
- Corresponding 32-bit words of the digest from consecutive message blocks are added to each other to form the digest of the whole message.
  - Automatic 32-bit word swapping to comply with the internal little-endian representation of the input bit-string
  - Supported word swapping format: bits, bytes, half-words, and 32-bit words
- Single 32-bit, write-only, input register associated with an internal input FIFO, corresponding to a 64-byte block size (16 × 32 bits)
- Automatic padding to complete the input bit string to fit the digest minimum block size
- AHB slave peripheral, accessible by 32-bit words only (else an AHB error is generated)
- 8 × 32-bit words (H0 to H15) for output message digest
- Automatic data flow control supporting direct memory access (DMA) using one channel
- Support for both single and fixed DMA burst transfers of four words
- Interruptible message digest computation, on a per-block basis
  - Reloadable digest registers
  - Hashing computation suspend/resume mechanism, including DMA